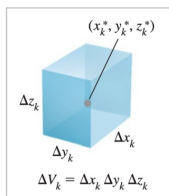
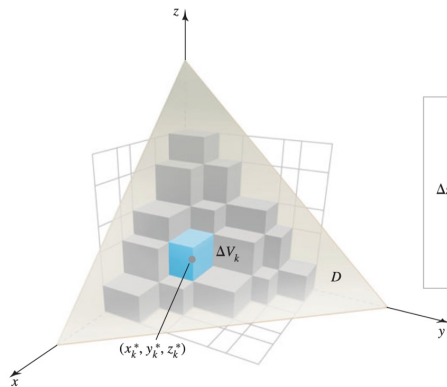


Lecture 13 Triple Integrals

Recap: Double integrals in polar coordinates

$$\iint_R f(x,y) dx dy = \iint_R f(r \cos \theta, r \sin \theta) r dr d\theta$$

Triple integrals: $\int_0^1 \int_0^1 \int_0^1 f(x,y,z) dx dy dz$



$$\iiint_D f(x,y,z) dx dy dz$$

$$\approx \lim_{\Delta \rightarrow 0} \sum f(x_k^*, y_k^*, z_k^*) \Delta x_k \Delta y_k \Delta z_k$$

$$\Delta = \max_{1 \leq k \leq n} \{x_k, y_k, z_k\}$$

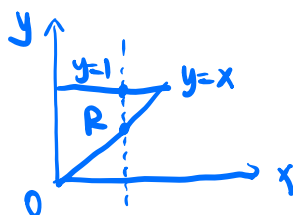
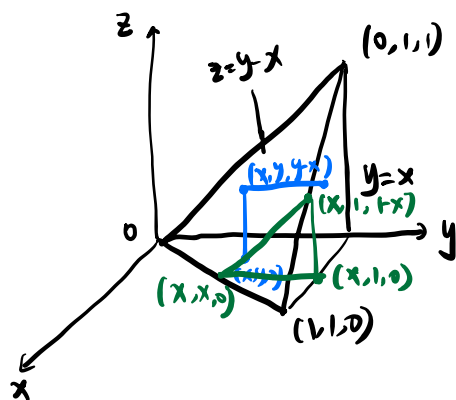
1° If $f(x,y,z) \equiv 1$, then $\iiint_D f(x,y,z) dx dy dz$ is the volume of the solid D

2° If $f(x,y,z)$ is the density at (x,y,z) , then $\iiint_D f(x,y,z) dx dy dz$ is the mass of the solid D .

$$\text{density} \times \text{volume} = \text{mass}$$

Example. Let D be the tetrahedral solid bounded by the plane $z = 1-x$, the plane $y=1$, the xy -plane and the yz -plane. Evaluate the triple integral

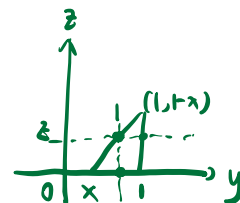
$$\iiint_D x^2 dz dy dx$$



Sol. Integration by pillars

$$\int_{x=0}^1 \int_{y=x}^1 \int_{z=0}^{z=y-x} x^2 dz dy dx$$

Integration by slices.



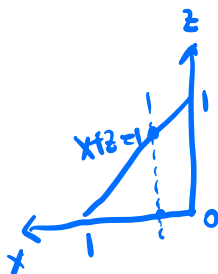
$$\begin{aligned} & \int_{x=0}^1 \int_{y=x}^1 \int_{z=0}^{z=y-x} x^2 dz dy dx \\ &= \int_0^1 \int_x^1 [x^2 z]_{z=0}^{z=y-x} dy dx \\ &= \int_0^1 \int_x^1 x^2 (y-x) dy dx \\ &= \int_0^1 \left[\frac{x^2 y^2}{2} - x^3 y \right]_{y=x}^{y=1} dx \\ &= \int_0^1 \left(\frac{x^2}{2} - x^3 - \frac{x^4}{2} + x^4 \right) dx \\ &= \int_0^1 \left(\frac{x^2}{2} - x^3 + \frac{x^4}{2} \right) dx \\ &= \left[\frac{x^3}{6} - \frac{x^4}{4} + \frac{x^5}{10} \right]_0^1 \\ &= \frac{1}{6} - \frac{1}{4} + \frac{1}{10} \\ &= \frac{1}{60} \end{aligned}$$

Integration by pillars

$$\int_{x=0}^1 \int_{z=0}^{z=1-x} \int_{y=x+z}^1 x^2 dy dz dx$$

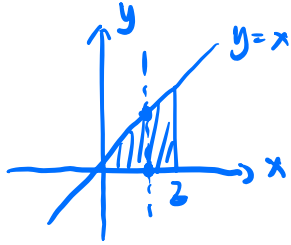
Integration by slices

$$\int_{x=0}^1 \int_{z=0}^{z=1-x} \int_{y=x+z}^1 x^2 dy dz dx$$



Example. Evaluate $\int_0^{\sqrt[4]{\pi}} \int_0^z \int_y^z 12y^2 z^3 \sin(x^4) dx dy dz$.

Sol. Fix z , $y \leq x \leq z$, $0 \leq y \leq z$



$$\int_0^{\sqrt[4]{\pi}} \int_0^z \int_y^z 12y^2 z^3 \sin(x^4) dx dy dz.$$

$$= \int_0^{\sqrt[4]{\pi}} \int_0^z \int_0^x 12y^2 z^3 \sin(x^4) dy dx dz.$$

$$= \int_0^{\sqrt[4]{\pi}} \int_0^z [4y^3 z^3 \sin(x^4)] \Big|_0^x dx dz$$

$$= \int_0^{\sqrt[4]{\pi}} \int_0^z 4x^3 z^3 \sin(x^4) dx dz$$

$$= \int_0^{\sqrt[4]{\pi}} \int_0^z z^3 \sin(x^4) d(x^4) dz$$

$$= \int_0^{\sqrt[4]{\pi}} z^3 (-\cos x^4) \Big|_0^z dz$$

$$= \int_0^{\sqrt[4]{\pi}} z^3 (1 - \cos z^4) dz.$$

$$= \int_0^{\sqrt[4]{\pi}} \frac{1 - \cos z^4}{4} d(z^4)$$

$$= \left[\frac{z^4}{4} - \frac{\sin z^4}{4} \right] \Big|_0^{\sqrt[4]{\pi}}$$

$$= \frac{\pi}{4}$$